

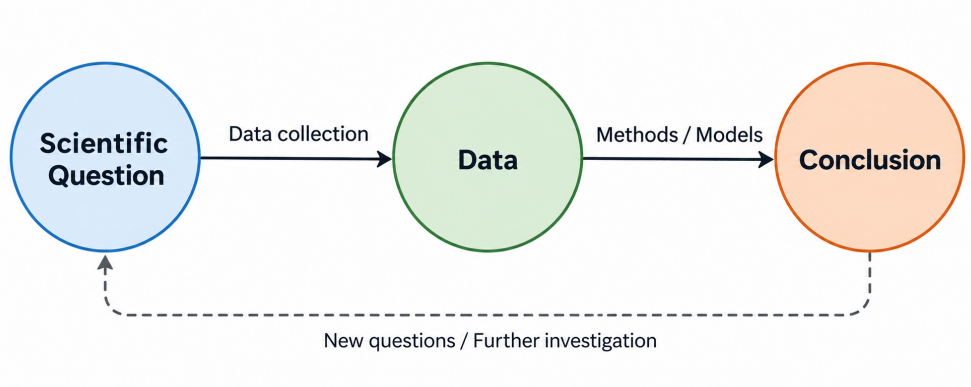
PH 724 Data Analysis & Scientific Communication Incorporating Use of Generative AI

Part 1: Introduction to Practice Project 1 (Rat Surgery, Chp. 1 of [Textbook])

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Big picture of statistics



Scientific context

- Diabetes is associated with complications following surgery, including impaired wound healing.
- Researchers investigated whether **hyperbaric oxygen treatment** could improve wound healing.

Scientific question

Does hyperbaric oxygen treatment improve the healing of surgical wounds in diabetic rats?

Study design

30 rats were made diabetic by administering a drug that destroys the pancreas.

All rats underwent surgery:

- An incision was made along the entire length of the back.
- The incision was closed with surgical staples.

The rats were then evenly divided into two groups:

Group	Treatment
Hyperbaric oxygen	100% oxygen, 2 atmospheres, 90 min/day
Control	Standard oxygen concentration and atmospheric pressure, 90 min/day

Final data

Rat	Site on the back: shoulder to tail					Mean
	A	B	C	D	E	
1 [†]	3.8300	7.3788	44.353	19.555	--	18.779
2	27.861	29.974	15.470	23.455	--	24.190
3	56.996	60.960	20.306	--	28.123	41.596
4	--	38.043	68.080	42.425	30.335	44.721
5	16.276	--	59.033	73.891	--	49.733
6	38.267	33.702	35.558	44.598	32.678	39.961
7	9.0384	11.259	27.121	31.984	--	19.851
8	16.728	27.590	13.238	12.139	6.3865	15.216
9	11.866	27.983	26.226	15.594	19.225	20.179
10	23.352	34.790	27.556	35.883	22.848	28.888
11	16.444	31.928	21.495	15.590	7.0750	18.506
12	23.342	46.313	33.810	15.686	--	29.788
13	15.267	14.452	10.635	22.156	6.8062	13.863
14	21.732	20.746	12.293	17.295	10.301	16.473
15 [†]	82.508	13.645	49.187	--	53.432	49.693
16	--	45.919	63.090	68.137	36.500	53.412
17	80.147	29.943	71.928	--	46.609	57.157
18	31.938	--	36.211	49.815	44.468	40.608
19	15.453	31.384	27.127	27.961	9.9035	22.366
20	21.183	27.429	20.058	--	--	22.890
21	20.445	12.532	15.661	28.694	--	19.333
22	16.928	59.579	29.407	18.626	8.8352	26.675
23	35.631	21.613	23.155	42.379	16.203	27.796
24	20.523	24.621	16.292	--	18.680	20.029
Mean	27.534	29.627	31.970	31.888	23.436	29.126

[†]Rats 1–14 are hyperbaric O₂-treated; 15–24 are the controls.
(Data published with permission from Dr. George Huang.)

- Multiple measurements are taken from each rat.
- Some site measurements are missing.
- **14** hyperbaric oxygen-treated rats and **10** control rats were included in the final analysis.
 - Six rats had glucose levels considered too low to be classified as diabetic. They were excluded from the main analysis.

Outcomes and explanatory variables

- Outcome: strength of skin specimens 24 days after surgery.
 - Higher strength indicates stronger healed tissue.
 - Some specimens could not be measured for procedural reasons, resulting in missing observations.
- Explanatory variables:
 - Treatment group
 - Site (5 locations along the back)

Assumptions

- The exclusion of the six rats is unrelated to treatment.
- Observations from different rats are independent.
- Measurements within the same rat may be correlated.
- (?) Missing site measurements are unrelated to skin strength.

Your task: preliminary analysis

Your preliminary work should include:

- Data cleaning
 - Check variable coding and missing values
- Exploratory data analysis
 - Examine the distribution of skin strength
 - Compare treatment groups
 - Examine differences across sites
 - Explore variation within and between rats
- Informative visualizations
 - Choose figures that help answer the scientific question and reveal the data structure
- A proposed plan for formal analysis
 - Consider the data type and assumptions
 - Be prepared to **justify your choices**